

What Rank-Based Reward Shaping Can Learn from Prediction Guarantees

Research phase: DRAFT after 1 round, 0 ITERATE, 0 REVISE. Edit phase: readable note

Abstract

Q surveys guarantees for algorithms that use fallible predictions, while P uses multimodal pairwise judgements to assign credit during cooperative agent training. The thread asked whether Q's account of prediction error and cost gives a guarantee for P's learned policy. The peer checks found exact cancellation of P's shaping signal under complete reward accounting, alongside conditional effects from pooling rewards or masking inactive agents. They also found that P's stated graph condition does not suffice for its unregularised score estimate. The verifier ended the thread as a draft because these results locate a useful test, but the reward routing and finite-training benefit in P remain unverified.

1 Introduction

Q is a survey of learning-augmented algorithms: methods that use predictions while stating what happens when those predictions are wrong. It separates a predictor's error from the cost of the decision made with it. It also asks how guarantees pass from one component to the next and how the cost of obtaining a prediction enters the account. A model's output needs a task-specific meaning and an error measure before an algorithmic guarantee can use it [7].

P addresses credit assignment in cooperative multi-agent learning, including teams whose active members change. Its MARS-RA method asks a large multimodal model which of two active agents contributed more, fits Bradley–Terry scores to the judgements, and turns the scores into reward-shaping potentials. P reports training outcomes and a trend in which more accurate comparisons and more queries accompany better performance. Its score-estimation claim concerns a latent preference of the comparator; its Shapley-value interpretation assumes that this preference matches the agents' true contributions [5].

The thread first asked whether Q's prediction contract could explain a downstream benefit in P. That would require a link from comparison error to learned-policy performance, together with the price of making comparisons and a rule for agents entering or leaving the team. The work narrowed to the point where that link would have to act: the shaped reward and the critic used during finite training.

2 What was done

The peers first checked P's score-estimation premise. They then traced one agent's potential through a whole episode and through a matched critic. An objection about changing teams forced them to repeat the calculation with an active-agent reward mask. They next checked what happens if the vector of shaped rewards is pooled into one team reward. Finally, they used a small policy-gradient example to test whether an accurate ranking is necessarily a useful training signal, and set these results against Q's requirements for downstream error and query cost.

P's stated protocol compares every ordered pair in the active set. If m_t agents are active at a queried step t , this calls the comparator $m_t(m_t - 1)$ times. P reports that, in its experimental setup, training for 50 million environment steps took 16 hours for MAPPO and 30 hours for MARS-RA. These figures motivate a charged-query comparison; they do not give a return-per-query guarantee [5]. Q's cost convention requires a declared conversion from calls to the objective being measured [7].

3 Results

What the potential does exactly

Let i denote one member of a fixed agent population, t a time step, T the end of an episode, and γ the discount factor. Write $\psi_{i,t}$ for that agent's potential at step t , using the same realised value on the two transitions that meet there. P's shaping increment is $F_{i,t} = \gamma\psi_{i,t+1} - \psi_{i,t}$. If every transition is credited and the terminal potential $\psi_{i,T}$ is zero, the peers checked the pathwise identity

$$\sum_{t=0}^{T-1} \gamma^t F_{i,t} = -\psi_{i,0} + \gamma^T \psi_{i,T} = -\psi_{i,0}.$$

The intermediate comparison scores cancel. When the initial potential is fixed before the first action, shaping shifts an exact episode return by a fixed amount and leaves its policy ordering unchanged. This is an application of existing potential-shaping results, including work on changing potentials and their relation to value initialisation [2, 1, 6].

The same cancellation reaches an exactly matched critic. Let ρ be P's shaping weight, $\widehat{V}_i^{\text{env}}$ an unshaped value estimate, and $\widehat{V}_i = \widehat{V}_i^{\text{env}} - \rho\psi_i$ its matched shaped version. The peers checked that their one-step temporal-difference residuals are equal. If the shaped critic has an additional error e_i , the shaped residual differs by $\gamma e_{i,t+1} - e_{i,t}$. Thus the comparison signal has no effect through an exactly matched residual, but critic mismatch gives a specific channel to measure during approximate training. This calculation does not show how P's implementation represents its critic.

What changes with a changing team

Complete accounting matters. Let $a_{i,t}$ equal one when agent i is active at t and zero otherwise. If the shaping increment is paid only when the agent is active, the peers obtained, with a zero terminal potential,

$$\sum_{t=0}^{T-1} \gamma^t a_{i,t} F_{i,t} = -a_{i,0} \psi_{i,0} + \sum_{t=1}^{T-1} \gamma^t (a_{i,t-1} - a_{i,t}) \psi_{i,t}.$$

On re-entry, the final sum can contain an uncompensated negative term. Setting the potential to zero while an agent is absent does not remove it. P gives a potential over the current

active set but does not specify enough about inactive-agent rewards and critic targets to determine whether its training uses this mask [5].

Pooling gives another conditional result. P forms potentials by taking a softmax of the active agents' scores, so their potentials sum to one whenever the active set is nonempty. If all agent coordinates, including zero coordinates for absent agents, are summed before a common team update, the pooled shaping increment depends only on this sum and carries no individual ranking information. Pooling only agents active at the start of a transition can retain a rank effect at entry, where the boundary accounting above also matters. Separate per-agent rewards can affect approximate updates. The peers established these alternatives from P's equations, but did not establish which route its implementation takes.

Why a better ranking is not enough

The peers also checked a one-step diagnostic. One agent chooses a binary action a , with each action equally likely; a second agent is inert, and the team return is $G = a$. For a logit policy, the score is $z = a - 1/2$. With a state-only baseline b , the gradient sample $g_b = z(G - b)$ has variance $(b - 1/2)^2/4$. The neutral baseline $b = 1/2$ therefore has zero variance, while the softmax of the rank-correct Shapley scores $(1/2, 0)$ gives the first agent $b > 1/2$ and positive variance. This tests whether ordinal accuracy alone predicts gradient quality; it is neither a MAPPO result nor a contradiction of P's reported experiments. Prior work already shows that potential shaping can raise policy-gradient variance [3]. Work on preference queries also identifies a gap between query choice and policy improvement [4].

4 Objections and their status

The first objection was to P's finite-sample Bradley–Terry claim. With two agents at a true tie and two independent comparisons, the comparison graph is connected. Both outcomes favour the same agent with probability $1/2$. On those samples an unregularised likelihood has no finite maximiser: the favoured score can keep moving away. This exceeds the failure probability $1/4$ allowed by P's stated $1 - n^{-2}$ claim when the number of agents n is two. The peers agreed on this counterexample. A constrained or regularised estimator and suitable sampling assumptions would be needed for a repaired result; the calculation does not dispute P's empirical outcomes [5].

The second objection was to a simpler cancellation argument. Zero potentials at absence and termination are insufficient when shaping rewards are masked for inactive agents. The active-mask identity above settled the mathematical point, while the implementation question remained open. The peers also noted that repeated model calls need not be independent, that P's comparison accuracy does not by itself control critic error, and that cancellation and variance effects have prior literature. No improved-return theorem or novelty claim follows from these checks.

5 Why the thread stopped

The verifier marked the thread DRAFT because it had reached a conditional result with a clear limit. Complete shaping cancels from exact returns; full pooling removes the ranking from a common reward; active-only masking can instead leave a re-entry term. These facts explain why Q's prediction-error framework cannot yet be transferred to P's learned-policy return. The smallest step to restart the work is to trace one entry or re-entry transition

through P's reward tensor, inactive-agent mask, and critic target. A matched-critic and neutral-potential control could then test whether any remaining effect improves held-out unshaped return under stated step, time, and query budgets.

References

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